Infertility & Assisted Reproductive Technique

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Out lines:

- Introduction.
- Definition of new terms.
- Risk factors of infertility
- Causes of infertility.
- Diagnosis of Infertility.
- Treatment of the Infertility.
- Nursing Role
- Assisted Reproductive

Learning Objectives:

At the end of this lecture students should be able to:

- Define each key term listed.
- List national and international Infertility prevalence.
- Mention Risk factors of infertility.
- Explain causes of infertility.
- List Diagnostic Studies of the Infertility.
- Explain treatment of the Infertility.

Definitions:

Infertility failure to achieve a clinical pregnancy after 12 months or more of regular unprotected sexual intercourse.

Sterility: Failure of conception due to incurable cause e.g., congenital absence of uterus

Type of infertility.

1. **Primary infertility:** Conception has never occurred.

2. **Secondary infertility**: failure of conception after a period of previous fertility

Incidence:

- Infertility affects Worldwide approximately 10-15% of reproductive-age couples
- In Egypt: 12%.

- Male factor: 30%
- Female factor: 60%
- Unexplained: 10%

Risk factors of infertility:

1. Advanced age
2. Excessive exercise
3. Smoking
4. Mental stress
5. Alcohol consumption
6. Being obese or overweight
7. Environmental/occupational factors
8. Toxic effects related to tobacco, marijuana, or other drugs

9. Inadequate diet associated with extreme weight loss or gain
10. Exposure to some chemicals * some pesticides, herbicides, metals (lead) and solvents.

Causes of infertility:

1-Female's causes include:

- **Cervical:** Stenosis or abnormalities of the mucus-sperm interaction
- **Uterine:** Congenital or acquired defects; may affect endometrium or myometrium; may be associated with primary infertility or with pregnancy wastage and premature delivery
- **Ovarian:** Alteration in the frequency and duration of the menstrual cycle—Failure to ovulate is the most common infertility problem, Ovulation problems (e.g. polycystic ovarian syndrome (PCOS).
- **Tubal:** Abnormalities or damage to the fallopian tube; may be congenital or acquired, Tubal blockage, previous tubal ligation.
- **Peritoneal:** Anatomic defects or physiologic dysfunctions (eg, infection, adhesions, adnexal masses)
- **Pelvic inflammatory disease** caused by infections like tuberculosis.
- Endometriosis.
- Infections such as chlamydia and/or scar tissue.
- Hyper prolactinemia.

2- Male's infertility causes include:

- Pretesticular: Congenital or acquired diseases of the hypothalamus, pituitary, or peripheral organs that alter the hypothalamic-pituitary axis
- Testicular: Genetic or nongenetic, malformations, hormone imbalance, or blockage of the man'sduct system.

- Posttesticular: Congenital or acquired factors that disrupt normal transport of sperm through the ductal system
- The main cause of male infertility is low semen quality.
- Low sperm count due to endocrine problems, drugs, radiation, or ifinfection.

3-Combined infertility:

In some cases, both the man and woman may be infertile or sub-fertile, and thecouple's infertility arises from the combination of these conditions as.

- Advanced age.
- Excessive exercise.
- Environmental/occupational factors.
- Toxic effects related to tobacco, marijuana, or other drugs.
- Inadequate diet associated with extreme weight loss or gain.

4-Unexplained infertility

- In these cases, abnormalities are likely to be present but not detected by current methods.
- Possible problems could be that
  - The egg is not released at the optimum time for fertilization that it may not enter the fallopian tube.
  - Sperm may not be able to reach the egg.
  - Fertilization may fail to occur.
  - Transport of the zygotemay be disturbed, or implantation fails.

Diagnosis of the Infertility

- Infertility is a problem that involves both partners. Diagnostic testing is unnecessary if the couple has not attempted to conceive for at least 1 year, unless the woman is age 35 years or older, or if they have a history of a male factor infertility, endometriosis, a tubal factor, diethylstilbestrol (DES) exposure, pelvic inflammatory disease, or pelvic surgery.
- A complete infertility evaluation is performed according to the woman's menstrual cycle and may take up to 2 menstrual cycles before the etiology is determined.

Obtain the following medical history and information from the couple:

- ✓ Copy of previous medical records.
- ✓ Completed medical history questionnaire.
- ✓ Details regarding the type of infertility (primary or secondary) and its duration.
- ✓ History of previous pregnancies and their outcomes; pregnancy intervals; and detailed information about pregnancy loss, pregnancy duration,
- ✓ History of previous infertility evaluation/treatment, including details about frequency of intercourse, use of lubricants, vaginal douches after intercourse, and presence of any sexual dysfunction.
- ✓ Female menstrual history, frequency, and patterns since menarche, as well as history of weight changes, hirsutism, frontal balding, and acne
- ✓ Male medical history, including previous semen analysis results, history of impotence, premature ejaculation, change in libido, history of testicular trauma, previous relationships, history of any previous pregnancy in female partners, and the existence of offspring from previous female partners
- ✓ Couple’s history of sexually transmitted diseases (STDs); surgical contraception (eg, vasectomy, tubal ligation); lifestyle; consumption of alcohol, tobacco, and recreational drugs (amount and frequency); occupation; and physical activities

- ✓ Couple's current medical treatment (if any), reason, and any history of allergies
- ✓ Complete review of systems to identify any endocrinologic or immunologic issue that may be associated with infertility

Examination for infertility should include the following:

- ✓ Routine records of blood pressure, pulse rate, and temperature (if applicable)
- ✓ Height/weight findings to calculate body mass index.
- ✓ Head and neck assessment: (1) The presence hyperthyroidism; (2) lower implantation of ears and hairline, and webbed neck can be associated with chromosomal abnormalities; (3) exclude thyroid gland enlargement/nodules, which may indicate thyroid dysfunction
- ✓ Breast evaluation: Assess breast development and seek any abnormal masses or secretions, especially galactorrhea
- ✓ Abdominal evaluation: Assess for presence of abnormal masses at hypogastrium level
- ✓ Gynecologic evaluation: Assess for hair distribution, clitoris size, Bartholin glands, labia majora/minora, and any condylomata acuminatum or other lesions that could indicate the existence of venereal disease
- ✓ Speculum examination: Obtain a Papanicolaou test and cultures for gonorrhea, chlamydia, Ureaplasma urealyticum,Mycoplasma hominis; assess for cervical stenosis
- ✓ Bimanual examination: Establish direction of the cervix plus size/position of the uterus to exclude the presence of uterine fibroids, adnexal masses, tenderness, or pelvic nodules indicative of infection or endometriosis; assess for defects (e.g, absence of vagina and uterus, vaginal septum).
- ✓ Extremities evaluation: Exclude malformation (eg, shortness of fourth finger, cubitus valgus), which can indicate chromosomal abnormalities and other congenital defects
- ✓ Dermatologic evaluation: Assess for the presence of acne, hypertrichosis, and hirsutism
- ✓ Urology examination of the male couple if the patient's history of his semen analysis produces an abnormal finding.

Attention should be directed to the following:

- Congenital abnormalities of the genital tract (eg, hypospadias, congenital absence of the vas deferens).
- Testicular size, urethral stenosis, and presence of any varicocele.
- Any previous inguinal hernia repair: Can indicate accidental ligation of spermatic artery

✓ Laboratory, imaging, and/or surgical evaluation

Laboratory, radiologic, and/or surgical assessment of the female includes the following:

- Cervical: Postcoital test or Sims-Huhner tests.
- Uterine and endometrial: Hysterosalpingogram—most frequently used diagnostic tool to assess endometrial cavity.
- Pelvic ultrasonograms; pelvic magnetic resonance imaging; hysteroscopy; endometrial biopsy, Hysterosalpingogram.

Tubal and peritoneal: Laparoscopy and hysterosalpingogram

- Ovarian: Progesterone levels and/or serial ultrasonography to assess ovulation; ovarian reserve

Laboratory evaluation of the male partner includes the following:

- Semen analysis: Volume, pH level, concentration, motility, morphology, WB count

- Sperm function tests:

(1) The acrosome reaction test with fluorescent antibodies.

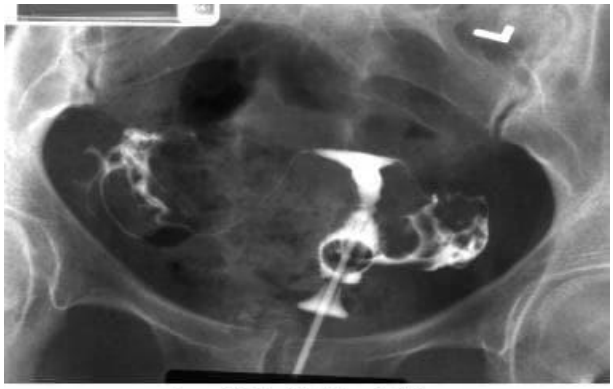
(2) Computer assessment of the sperm head.

(3) Computer motility assessment.

(4) hemizona-binding assay.

(5) Hamster penetration test,.

(6) Human sperm-zona penetration assay.



Normal HSG with Bilateral Spillage

Diagnostic Studies for Females:

- Serum Hormone Testing. FSH, LH
- Postcoital Exam
  - Couple has sexual intercourse 8 – 12 hours prior to exam, 1-2 days before expected ovulation.
  - A 10-cc syringe with catheter attached is used to collect a specimen of secretions from the vagina.
  - Secretions are examined for: S/S infection, # of active & non-motile sperm, sperm-mucus interactions, & consistency of cervical mucus.
- Endometrial Biopsy.
- Hysterosalpingogram (HSG)
  - Detects uterine anomalies (septate, unicornate, bicornate)
  - Detects Tubal anomalies or blockage
  - Iodine-based radio-opaque dye is instilled through a catheter into the uterus and tubes to outline these structures and x-rays are taken to document findings.
- Laparoscopy
  - General or epidural anesthesia
  - Abdomen is insufflated with carbon dioxide,
  - One or more trochars are inserted into the peritoneum near the umbilicus & symphysis pubis
  - Laparoscope visualizes structures in the pelvis
  - Can perform certain surgical procedures.

Diagnostic Studies of the Males:

- Semen analysis
  - Ejaculates into a specimen container

- Ejaculate examined for:
  - Number
  - Morphology
  - Motility

Normal Semen Analysis Results

- Volume >2.0 mL
- pH 7.0 – 8.0
- Total sperm count >20 million per mL
- Motility 50% or greater
- Normal forms 50% or greater.

Treatment of Infertility

- Treatment plans are based on the diagnosis, duration of infertility, and the woman's age. Management of any underlying female and/or male factors affecting fertility may include
- Medical treatment (eg, pharmacotherapy),
- Surgical intervention or both.
- Assisted conception

Treatment of Female Infertility

Primary

- Change environment
- Timing intercourse - checking for ovulation.
- Decrease stress, diet, exercise, etc

Secondary

- Correct hormone levels,

- **Medicines** to help the ovaries release an egg if ovulation is irregular or stops happening.

Common fertility medicines include:

- Clomifene – encourages the monthly release of an egg (ovulation).
- Tamoxifen – an alternative to clomifene that may be offered if you have ovulation problems
- Metformin is particularly beneficial for women who have polycystic ovary syndrome (PCOS)
- Gonadotrophins – can help stimulate ovulation in women, and may also improve fertility in men
- Gonadotrophin-releasing hormone and dopamine agonists –
- Other types of medicine prescribed to encourage ovulation in women.

✓ Medicine that stimulates the ovaries is not recommended for women with unexplained infertility because it has not been found to increase their chances of getting pregnant.

Surgery

There are several types of surgical procedures that may be used to investigate fertility problems and help with fertility.

- Correct physical problems include polyps, scar tissue and some fibroids.
- Surgery can be used to break up the scar tissue in your fallopian tubes, making it easier for eggs to pass through them.
- Laparoscopic surgery with small cuts or traditional surgery with a large cut in the stomach area may be needed to treat conditions such as

endometriosis, pelvic adhesions and larger fibroids.

- Final – Assisted Reproduction Technologies, more often necessary for female infertility.

Treatment of Male Infertility

Primary –

- Change environment and lifestyle as.
- Avoid heat, drugs, toxins, and timing intercourse.
- Regular exercise and drink less alcohol or tobacco.

Secondary –

- Correct hormone levels.
- Medicines – to improve sperm count by help the testicles function better too.

Surgery –

- Corrects physical problems, immune suppressants, repairing a large varicocele.
- Correcting an epididymal blockage and surgery to retrieve sperm

Sometimes the epididymis becomes blocked, preventing sperm from being ejaculated normally. If this is causing infertility, surgery can be used to correct the blockage.

- Sperm retrieval procedures**.** These techniques can collect sperm no ejaculate, or if no sperm is in your semen. Sperm retrieval procedures also may be used when assisted reproductive techniques are planned and sperm counts are low or irregular.

- Final – Assisted Reproduction Technologies.

Assisted Reproductive Technique

Assisted reproductive technologies used to treat infertility include the following:

- Artificial Insemination
- In vitro fertilization (IVF)
- Gamete intrafallopian transfer (GIFT)
- Zygote intrafallopian transfer ZIFT)
- Intracytoplasmic sperm injection (ICSI)
- Intrauterine insemination (IUI)
- Sperm, oocyte, or embryo cryopreservation

1. Artificial Insemination

Indications

- Failure of deposition of semen I.e., hypospadias, impotence, vaginismus, premature ejaculation

- Hostile cervical mucus -Oligospermia

Technique

- Induction of ovulation
- After 12-36 hours later 0,5ml of processed semen is injected inside the uterus using a special catheter.
- Injection may be intracervical, intrauterine, intra-tubal, or intrafollicular.

2. In-vitro fertilization and embryo transfer (IVF and ET

Indications

- Tubal factor
- Unexplained infertility.
- Infertility with endometriosis
- Oligospermia of the husband

Technique- :

- Induction of ovulation
- Aspiration of the ova 32h after induction of ovulation under local or general anesthesia.
- Fertilization is done by adding prepared sperms to aspirated ova in a culturemedia.
- Embryo transfer...fertilized eggs are transferred into the uterus using Trans cervical catheter.

3. Gamete intra fallopian transfer...GIFI

Indications

- Abnormal semen
- Endometriosis

- Hostile cervical mucus
- Failure of IVF.

Technique- :

- Induction of ovulation and aspiration of ova as IVF
- Injection of ova mixed with prepared sperms into the ampullary portion of the tube using laparoscopy

4. Zygote intra fallopian transferee

- As GIFT but zygotes are injected into the tube
- It is not widely used.

5. Sub zonal injection of sperms (SUZI(

Indications

- Failure of in vitro fertilization

Technique

- A hole is made in zona pellucida, and sperms are introduced to sub zonal space.

6. Intra cytoplasmic sperm injection ...ICSI

Indications

- Failure of in vitro fertilization for 3 cycles
- Marked oligospermia <80,000 sperms in the specimen.
- Anti-sperm antibodies
- Immotile sperms but still alive

Technique

Under microscope, single sperm is injected into cytoplasm of the oocyte using micropipette.

Effects of Infertility

Psychological impact

- Partners may become more anxious to conceive, increasing sexual dysfunction.
- Marital discord often develops in infertile couples, especially when they are under pressure to make medical decisions.
- Women trying to conceive often have clinical depression rates and feeling a loss of control over one‘s life; similar to women who have heart disease or cancer.
- Even couples undertaking IVF face considerable stress.

Social impact

Groups, a degree of rejection (or a sense of being rejected by the couple) maycause considerable anxiety and disappointment.

Role of Nurse in Infertility counseling

1. Sending initial information to couples who have enquired about treatment.
2. Reviewing the client‘s medical records to plan treatment cycle dates
3. Once the cycle plan has been confirmed, ensuring that the patient is telephoned, and the treatment plan is sent to couples to confirm

treatment dates.

4. Making appointments in accordance with the treatment plans and clinic procedures.
5. Providing support for the couple as they undergo diagnosis and their chosen treatment option.
6. .Therapeutic communication skills are an essential component in nursing care.
7. Receiving the patient & family and make them accessible & comfortable for counseling.
8. Fertility nurse specialists provide care for the individuals and couples before, during, and after fertility treatment.
9. The nurse should reinforce the couple's coping strategies or assist them in exploring adoptive alternatives to poor coping, increasing their sense of control in this situation.
10. Evaluating the need for referrals to other resources, such as psychological or pastoral counseling.
11. The nurse should instruct and teach the women with the following points.